Department of Education

STEM Education
National Schools of Excellence
Chemistry Syllabus
Grade 12
Term 1

DOMAIN	Phylum
SCIENCE	Chemistry
Technology	
Engineering	
Mathematics	

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SECRETARY’S MESSAGE

This Chemistry Syllabus is to be used exclusively by teachers of Chemistry to teach Grade 12 Students enrolled for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The Grade 11 STEM Chemistry Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Chemistry before undertaking the advanced concepts introduced in this syllabus.

This syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principal object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and Skills and apply them to solve every day problems whilst appreciating Chemistry as a fundamental field of science.

This STEM Chemistry Syllabus for Grade 12 constitutes six out of the ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Chemistry.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Chemistry is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Chemistry to human lives, including good health and environment.

In order to ensure that teachers discharge their responsibilities adequately this syllabus is accompanied by a Teacher’s Guide and Teacher’s Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for Chemistry to be used for the Grade 12 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

Chemistry is the study of matter and its composition, how they interact to form new compounds through chemical reactions and the energy they generate as a result of these interactions. The concepts taught in Chemistry help us to understand these interactions at the molecular and atomic levels. Every substance, whether naturally occurring or artificially produced, consists of one or more elements of the periodic table.

Since Chemistry is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Chemistry.

The Grade 12 Chemistry Syllabus entails six (6) of ten (1) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

Since Chemistry is a highly specialized subject students undertaking Chemistry are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of Chemistry through the process of imparting basic chemistry knowledge and skills to students.

THEMA	MODULE	TOPICS
BIOTECHNOLOGY	12.1 Protein Sequencing	1. Amino Acids 2. Protein Structure and Function 3. Protein Purification 4. Protein Sequencing
HEALTH AND MEDICINE	12.1 Drug Development Process	1. Drug Targets 2. Drug Discovery 3. Drug Development and Production
ENERGY	12.1 Hydrogen – Future of Energy	1. Hydrogen as Fuel 2. Methods of Producing Hydrogen 3. Storing Hydrogen 4. Hydrogen Fuel Cell
BATTERY TECHNOLOGY	12.1 Solid-State Battery	1. Introduction to Solid-State Battery 2. Solid Electrolyte 3. Manufacturing process Solid-State Battery
ARTIFICIAL INTELLIGENCE	12.1 Artificial Molecular Machines	1. Introduction to Artificial molecular Machines 2. Catenanes and Rotaxanes 3. Other Protein based Motors under development
ROBOTICS	12.1 Robotics Meets Chemistry	1. Acidity Theories 2. Robotics 3. Programming a Robot

Since Chemistry is a highly specialized subject students undertaking Chemistry are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language Skills.

The modules in this Syllabus are sequenced to guide teachers of Chemistry through the process of imparting basic Chemistry knowledge and Skills to students.

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honourable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;

2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;

3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and

4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In the STEM Grade 11 Chemistry Syllabus, students were introduced to the basic concepts and knowledge of Chemistry in order for them to develop an appreciation of the discipline in a broader context. In this syllabus students are introduced to advanced concepts in Chemistry as they are applied to six Thema.

The introduction of the STEM Curriculum will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

NEW NOMENCLATURE

As part of the overall Curriculum Development, new nomenclature was introduced. These are explained below:

Term	Explanation
Domain	Each STEM area is defined as *domain* (e.g., Science Domain, Technology Domain, etc.)
Phylum	What would normally be regarded as “subject” (e.g., Chemistry, Physics, Chemistry, etc.) is now regarded as *phylum*; the plural is *phyla*.
Thema	Thematic areas of studies
Module	A specific concept within the Thema which is being studied
Topic	A specific aspect of the Module that is being studied

STEM Domains

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science
It is the study of the natural world, including the laws of the nature associated with physics, chemistry and Chemistry and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Chemistry, Chemistry and Physics.

Technology
Comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering
Concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics
It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM GRADE 12 CHEMISTRY CONTENT OVERVIEW – Term 1

Table 1.1 Sequencing & Benchmarking STEM Chemistry Content Standards

Setting STEM Chemistry Content Standards for Progressive Learning & Assessment			
Thema Biotechnology	**Module Content Standards**		
Module 12.1 Protein Sequencing	**12.1:** To determine the amino acid sequence of a protein using various purification methods and sequencing methods.		
Topics	**Performance Indicators Students are expected to:**	**Module Content Benchmark**	**Assessment Tasks**
1. Amino Acids	**12.1.1a:** Explain what amino acids are. **12.1.1b:** Draw and describe the chemical structure of an alpha-amino acid. **12.1.1c:** Name all 20 amino acids and identify them using the one-letter code and three-letter code. **12.1.1d:** Classify the 20 amino acids into Polar, Non-polar, Basic and Acidic. **12.1.1e:** Explain what peptide bond is. **12.1.1f:** Write the condensation reaction of two amino acids to form a dipeptide. **12.1.1g:** Differentiate between the N-terminus and the C-terminus.		
2. Protein Structure and Function	**12.1.2a:** Explain the four organizational levels of Protein Structure. **12.1.2b:** Differentiate between Polypeptides and Proteins. **12.1.2c:** Explain how chaperonins help in protein folding.		

	12.1.2d: Explain what Denaturation of Protein is. **12.1.2d:** Justify how the structure of a protein determines the function of that protein.		
3. Protein Purification	**12.1.3a:** List and explains the different ways in which to purify proteins. **12.1.3b:** Explain why purification of proteins is an important step. **12.1.3c:** Describe the process of protein purification using Electrophoresis. **12.1.3d:** Purify a Protein using any one of those methods.		
4. Protein Sequencing	**12.1.4a:** Define Protein Sequencing. **12.1.4b:** Explain the Process of Sanger's Method. **12.1.4c:** Explain the Process of Edman Degradation Method. **12.1.4d:** Utilize any one the protein sequencing methods to sequence a protein sample.		

Thema Health and Medicine	**Module Content Standards**		
Module 12.1 Drug Development Process	**12.1:** To present a plan on how one would take on drug discover, design and development process for a drug idea he/she has.		
Topics	**Performance Indicators Students are expected to:**	**Module Content Benchmark**	**Assessment Tasks**
1. Drug Targets	**12.1.1a:** Explain what drugs are and their importance. **12.1.1b:** List and discuss the different sources of drugs and lead compounds. **12.1.1c:** Explain the routes of drug administration. **12.1.1d:** Describe both the Pharmacokinetic phase and the Pharmacodynamic phase.	Students' ability to master the knowledge and Skills of: • Drug Discovery Process • Drug Design using Computer Aided Design Programs • Drug Development Process and Patents.	
2. Drug Discovery	**12.1.2a:** Explain the Process of Drug Discover. **12.1.2b:** Justify why Water Solubility is important. **12.1.2c:** Discuss what salt formation is. **12.1.2d:** Describe the process of incorporating water solubilizing groups in a drug structure.		
3. Drug Development and Production	**12.1.3a:** Explain the function of antibacterial agents. **12.1.3b:** Explain the function of antiviral agents. **12.1.3c:** Explain the function of anticancer agents. **12.1.3d:** Explain the function of cholinergic, anticholinergics and anticholinesterases. **12.1.3e:** Explain how drugs act on the Adrenergic Nervous System. **12.1.3f:** Explain the function of Opioid Analgesics. **12.1.3g:** Explain the function of antiulcer agents.		

Table 1.2 Mapping Content Progressions by Modules

Thema/Module	Module Content Standards	12.1	12.1
Biotechnology / Module 12.1: Protein Sequencing	**12.1:** To determine the amino acid sequence of a protein using various purification methods and sequencing methods.	✓	
Health and Medicine / Module 12.1: Drug Development Process	**12.1:** To present a plan on how one would take on drug discover, design and development process for a drug idea he/she has.		✓

THEMA BIOTECHNOLOGY

Introduction

Biotechnology is a branch of biology and chemistry that employs mathematics and physics to develop tools and engineering insights for modern biology and biomedical research. It entails the study and application of living cells and cellular materials in the development of pharmacological, diagnostic, agricultural, environmental, and other products that promote human welfare and society. It's also utilized to examine and manipulate genetic information in animals in order to imitate and study human disease. Biotechnology is a science that is both fundamental and applied. Agriculture, health care and medical sciences, biomedical engineering, cell biology, immunology, seed technology, virology, plant physiology, forensics, industrial processing, and environmental management all benefit from the technologies it develops.

There are a number of career paths linked to the field of Biotechnology, such as: working in research labs, graduate training in biology and chemistry, and professional careers in the health sciences. If you are someone who enjoys STEM (particularly biology and chemistry), below are some best biotechnology careers that may be of interest to you (research on them for more information)

- Biomedical Engineer
- Biochemistry
- Medical Scientist
- Clinical Technician
- Microbiologist
- Process Development Scientist
- Biomanufacturing Specialist
- Business Development Manager
- Product Strategist
- Biopharma Sales Representative
- Medical Scientist
- Biotechnological Technician
- Epidemiologist
- Microbiologist
- Medical and Clinical Lab Technologist
- Biomanufacturing Specialist
- Bioproduction Specialist

MODULE 12.1: PROTEIN SEQUENCING

4-5 Weeks

Protein sequencing is the technique of determining the amino acid sequence of a protein or peptide in its entirety or in part.[1] This module is designed to introduced students to the process or steps involved in Protein sequencing. The process involves:

1. Separating proteins into subunits
2. Producing Smaller Peptide Fragments
3. Sequencing the Protein Fragments
4. Repeating steps 2 and 3 using a different cleavage/hydrolysis method
5. Assembling the overlapping fragments for the full protein sequence

The entire practical learning experience will be covered within the following topics:
Topic 1: Amino Acids
Topic 2: Protein – Structure and Function
Topic 3: Protein Purification
Topic 4: Protein Sequencing

Linking Modules

- STEM-proper Grade 12 / Biology Phylum / Thema: Biotechnology / Module 12.1: Bioinformatics
- STEM-proper Grade 11 / Chemistry Phylum / Thema: Biotechnology / Module 11.1: Principles of Biotechnology / Topic 3: Proteins

Module Content Standard

12.1: To determine the amino acid sequence of a protein using various purification methods and sequencing methods.

Module Content Benchmark

Students' ability to master the knowledge and Skills of:

- Protein Sequencing Process
- Protein Purification Methods
- Protein Sequencing Methods

[1] (Protein Sequencing, 2021)

Topic 1: Amino Acids

Performance Indicators

Students are expected to:

- **12.1.1a:** Explain what amino acids are.
- **12.1.1b:** Draw and describe the chemical structure of an alpha-amino acid.
- **12.1.1c:** Name all 20 amino acids and identify them using the one-letter code and three-letter code.
- **12.1.1d:** Classify the 20 amino acids into Polar, Non-polar, Basic and Acidic.
- **12.1.1e:** Explain what peptide bond is.
- **12.1.1f:** Write the condensation reaction of two amino acids to form a dipeptide.
- **12.1.1g:** Differentiate between the N-terminus and the C-terminus.

Content Expansion

Body of Knowledge for Understanding

- Overview
- Protein: What are They?
- The alpha α-amino acid Structure
- 20 common amino acid
- Classification of the 20 amino acids
    - Side Chains with Basic Groups
    - Side Chains with Acidic Groups
    - Side Chains with Polar but Uncharged Groups
    - Side Chains with Nonpolar Groups
- Peptide Bonds
- N-Terminus vs. C-Terminus

Set of Skills for Proficiency Levels

- Analytical problem solving
- Research Skills
- Numerical Skills
- Written and Oral Communication Skills
- Observational Skills
- Planning Skills
- Teamwork and interpersonal Skills

Traits of Attitudes and Values to govern their lives

- Patience
- Determination
- Flexibility
- A logical and independent mind
- Meticulous attention to detail and accuracy

Topic 2: Protein Structure and Function

Performance Indicators
Students are expected to:

12.1.2a: Explain the four organizational levels of Protein Structure
12.1.2b: Differentiate between Polypeptides and Proteins
12.1.2c: Explain how chaperonins help in protein folding
12.1.2d: Explain what Denaturation of Protein is
12.1.2d: Justify how the structure of a protein determines the function of that protein

Content Expansion
Body of Knowledge for Understanding[2]

- Primary Structure
- Secondary Structure
- Tertiary Structure
- Quaternary Structure
- Polypeptide vs. Protein
- Chaperonins
- Denaturation
- Structure-Function Relationship

Set of Skills for Proficiency Levels

- Analytical problem solving
- Research Skills
- Numerical Skills
- Written and Oral Communication Skills
- Observational Skills
- Planning Skills
- Teamwork and interpersonal Skills

Traits of Attitudes and Values to govern their lives

- Patience
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[2] https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fcourses.mak.ac.ug%2Fsites%2Fdefault%2Ffiles%2Fdownloads%2FIndustrial%2520Biochemistry.doc&wdOrigin=BROWSELINK

Topic 3: Protein Purification

Performance Indicators

Students are expected to:

12.1.3a: List and explains the different ways in which to purify proteins.
12.1.3b: Explain why purification of proteins is an important step.
12.1.3c: Describe the process of protein purification using Electrophoresis.
12.1.3d: Purify a Protein using any one of those methods.

Content Expansion

Body of Knowledge for Understanding

- Overview
  - What is Protein Purification
- How does Protein Purification Work?
  - Cell Lysis
  - Clarification
  - Protein Binding
  - Elution
- Protein Purification Methods and Strategies
  - Column Chromatography
  - Ion-Exchange Chromatography
  - Reverse Phase Chromatography
  - Gel Filtration
  - Affinity Chromatography
  - Proteins and Other Charged Biological Polymers Migrate in an Electric Field
  - Protein Electrophoresis

Set of Skills for Proficiency Levels

- Analytical problem solving
- Research Skills
- Numerical Skills
- Written and Oral Communication Skills
- Observational Skills
- Teamwork and interpersonal Skills

Traits of Attitudes and Values to govern their lives

- Patience
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- Flexibility
- A logical and independent mind
- Meticulous attention to detail and accuracy

Topic 4: Protein Sequencing

Performance Indicators

Students are expected to:

12.1.4a: Define Protein Sequencing.
12.1.4b: Explain the Process of Sanger's Method.
12.1.4c: Explain the Process of Edman Degradation Method.
12.1.4d: Utilize any one the protein sequencing methods to sequence a protein sample.

Content Expansion

Body of Knowledge for Understanding [3]

- What is Protein Sequencing
- Recall: How Do Amino Acids Form Protein?
- Sanger's Method: Finding the Terminals
    - Unravelling the Protein
- Edman Degradation Method: Arriving at Different Solutions
- More Advanced methods
    - Mass Spectrometry
    - Dansyl Chloride Method

Set of Skills for Proficiency Levels

- Analytical problem solving
- Research Skills
- Numerical Skills
- Written and Oral Communication Skills
- Observational Skills
- Planning Skills
- Teamwork and interpersonal Skills

Traits of Attitudes and Values to govern their lives

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[3] https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fcourses.mak.ac.ug%2Fsites%2Fdefault%2Ffiles%2Fdownloads%2FIndustrial%2520Biochemistry.doc&wdOrigin=BROWSELINK

THEMA HEALTH AND MEDICINE

Introduction

Medicine is the art, science and practice of caring for a patient and managing the diagnosis, prognosis, prevention, treatment or palliative care of their injury or diseases. Medicine encompasses health care practices evolved to maintain and restore health by the prevention and treatment of illness. The practice of medicine has significantly changed with the advent of molecular biology with its enormous implications in biological sciences (sequencing of human genome), sophisticated new imaging techniques and advances in bioinformatics and information technology. That has sparked an explosion of scientific information that fundamentally influenced and changed the way in which patient care and management has been practiced.

The knowledge gleaned from the science of medicine has already improved and undoubtedly will further improve our understanding of complex disease processes and provide new approaches to disease treatment and prevention. Though the practice of medicine requires a scientific basis, the combination of medical knowledge, intuition, experience and judgment defines the art of medicine.

The study of the human and body, how this function, and how they interact, not only with each other but also with their environment, has been of utmost importance in ensuring human well-being. Research on potential treatments and preventive medicine has expended greatly with the development of modern medicine, and a network of disciplines, including such fields as genetics, psychology, and nutrition, aims to facilitate the betterment of our health.

MODULE 12.1: DRUG DEVELOPMENT PROCESS

4-5 Weeks

Papua New Guinea (PNG) has the potential to solve some of the world's health-related problems regarding therapy and treatments drugs. Why? Because PNG has majorly unexplored rainforests and large biodiversity. Think about it! PNG has a rich tradition of using herbal medicine, and about 50% of our rural population relies on traditional herbal medicine for therapy and treatment. This should imply to us that PNG is a potential source for drug discovery[4].

We (Students) need to be encouraged to exploit the scientific opportunities that our country has. If we choose to investigate that comparative advantage, we may likely stumble upon what may be new medical technologies or drugs that may not only solve PNG's health-related problems, but the worlds.

Medicinal Chemistry is an interdisciplinary study that combines aspects of chemistry, physical chemistry, pharmacology, microbiology, biochemistry, and computational chemistry. Medicinal Chemistry to the micromolecular level is a very interesting yet complex topic. However, for this Module, the focus will be on the process of discovery, design and development of drugs at a bird-eye view. By the end of this Module, the students should be sufficiently equipped with the concept of drug discovery, design and development should they choose to conduct further research on drug development as well as pursue a career in the field of pharmaceutical biotechnology.

Linking Modules

Prerequisites and links:

- Biology / Thema: Health and Medicine / Module 11.1: Health and Medicine Sciences
- Chemistry / Thema: Health and Medicine / Module 11.1: Biochemistry in Medicine

Module Content Standard

12.1: To present a plan on how one would take on drug discover, design and development process for a drug idea he/she has.

Module Content Benchmark

Students' ability to master the knowledge and Skills of:

- Drug Discovery Process
- Drug Design using Computer Aided Design Programs
- Drug Development Process and Patents.

[4] 2327-5162.S1.011-006.pdf (hilarispublisher.com)

Topic 1: Drug Targets

Performance Indicators

Students are expected to:

12.1.1a: Explain what drugs are and their importance.
12.1.1b: List and discuss the different sources of drugs and lead compounds.
12.1.1c: Explain the routes of drug administration.
12.1.1d: Describe both the Pharmacokinetic phase and the Pharmacodynamic phase.

Content Expansion

Body of Knowledge for Understanding

- Introduction
- Drugs: What are they?
- Drugs: Why do we need new ones?
- Drug Discovery and Design, a historical outline
- Sources of Drugs and Lead Compounds
    - Natural Resources
    - Drug Synthesis
    - Market forces and “me-too drugs”
- Classification of Drugs
- Routs of administration (Pharmaceutical Phase)
- Drug Action
    - Pharmacokinetic phase
        - Absorption
        - Distribution
        - Metabolism
        - Elimination
        - Bioactivity of drug
    - Pharmacodynamic Phase

Set of Skills for Proficiency Levels

- Analytical problem solving
- Research Skills
- Numerical Skills
- Written and Oral Communication Skills
- Observational Skills
- Planning Skills
- Teamwork and interpersonal Skills

Traits of Attitudes and Values to govern their lives

- Patience
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Topic 2: Drug Discovery

Performance Indicators

Students are expected to:

12.1.2a: Explain the Process of Drug Discover.
12.1.2b: Justify why Water Solubility is important.
12.1.2c: Discuss what salt formation is.
12.1.2d: Describe the process of incorporating water solubilizing groups in a drug structure.

Content Expansion

Body of Knowledge for Understanding

- Introduction
- Stereochemistry and Drug Design
    - Structurally rigid groups
    - Conformation
    - Configuration
- Solubility and Drug Design
    - Importance of water solubility
- Solubility and Drug Structure
    - Salt Formation
- Incorporation of Water Solubilizing Groups in a Structure
    - Type of Group
    - Reversibly and irreversibly attached groups
    - The position of the water solubilizing group
    - Methods of Introduction

Set of Skills for Proficiency Levels

- Analytical problem solving
- Research Skills
- Numerical Skills
- Written and Oral Communication Skills
- Observational Skills
- Planning Skills
- Teamwork and interpersonal Skills

Traits of Attitudes and Values to govern their lives

- Patience
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- Meticulous attention to detail and accuracy

Topic 3: Drug Development and Production

Performance Indicators

Students are expected to:

12.1.3a: Explain the function of antibacterial agents.
12.1.3b: Explain the function of antiviral agents.
12.1.3c: Explain the function of anticancer agents.
12.1.3d: Explain the function of cholinergic, anticholinergics and anticholinesterases.
12.1.3e: Explain how drugs act on the Adrenergic Nervous System.
12.1.3f: Explain the function of Opioid Analgesics.
12.1.3g: Explain the function of antiulcer agents.

Content Expansion

Body of Knowledge for Understanding

- Introduction
- Chemical Development
    - Chemical Engineering Issue
    - Chemical Plant, Health and Safety Considerations
    - Synthesis Quality Control
    - A Case Study
- Pharmacological and Toxicological Testing
- Drug Metabolism and Pharmacokinetics
    - Formulation Development
- Production and Quality Control
- Patent Protection
- Regulation

Set of Skills for Proficiency Levels

- Analytical problem solving
- Research Skills
- Numerical Skills
- Written and Oral Communication Skills
- Observational Skills
- Planning Skills
- Teamwork and interpersonal Skills

Traits of Attitudes and Values to govern their lives

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REFERENCES

- Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de)
- Robot Programming: A Guide to Programming Your Own Robots (edgy.app)

- Nanorobotics (fractal.org)
- Making and Operating Molecular Machines: A Multidisciplinary Challenge - Baroncini - 2018 - ChemistryOpen - Wiley Online Library
- Driving Smart Molecular Systems by Artificial Molecular Machines (wiley.com)
- Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de)
- https://www.scienceinschool.org/article/2016/chembot-chemistry-robots/
- Robot Programming: A Guide to Programming Your Own Robots (edgy.app)
- Robotics Designing of PCB - javatpoint
- https://blog.finology.in/Legal-news/Ways-to-Commercialize-IPR
- https://www.upcounsel.com/why-do-patents-expire
- Ihttps://courses.lumenlearning.com/wmopen-introductiontobusiness/chapter/intellectual-property-2/
- https://www.forbes.com/sites/arielcohen/2021/02/11/what-batteries-will-power-the-future/?sh=5ce1b64841c0
- https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fcourses.mak.ac.ug%2Fsites%2Fdefault%2Ffiles%2Fdownloads%2FIndustrial%2520Biochemistry.doc&wdOrigin=BROWSELINK
- https://www.biochemistry.org/education/careers/becoming-a-bioscientist/what-is-biochemistry/
- https://chemintelligence.com/ai-for-chemistry#:~:text=Artificial%20Intelligence%20%28AI%29%20is%20being%20used%20more%20and,and%20the%20time%20to%20market%20for%20new%20drugs.
- https://www.javatpoint.com/facebook-data-mining